Applied Statistical Modeling 2

Lorenzoni Valentina

Institute of Management - Scuola Superiore Sant' Anna, Pisa

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Tobit regression

The reasoning for the standard Tobit model could be extended to more general cases of censored variables

$$f(x;\mu,\sigma,c) = \begin{cases} \Phi\left(\frac{c-\mu}{\sigma}\right) & \text{if } y = c \\ \frac{1}{\sigma}\phi\left(\frac{y-\mu}{\sigma}\right) & \text{if } y > c \end{cases}$$

Zero Inflated models

For each of the zero-inflated models, an observed variable Y is assumed to be generated by two underlying processes

- a Bernoulli process determining wheter y_i is a **structural zero**, or a count (including some zeroes)
- a Poisson, Negative Binomial or Beta process generating counts (including some zeroes)

Two-part models

Two-part models could represent another approach to deal with the "problem of many zeroes", in this approach, data are thought to be generated from two processes

- one determining whether the outcome is zero (the binary part of the model), and
- the other determining the actual value if it is nonzero (the count/continuous part of the model)

Two-part models

In the two-part model,

- a binary choice model (i.e., probit, logit) is fit for the probability of observing a positive-versus-zero outcome, and
- conditional on a positive value, an appropriate regression model (i.e., Poisson, Gamma) is fit for the positive outcome
- we assume two different processes not a unique "unobserved" latent variable as for Tobit

Two-part models

Two-part models are sometimes referred to as "generalized" tobit models, and there are conceptual differences between the two-part and zero-inflated models, in the latter

- zero are "real" zero and could not be distinguished between "structural" and "real" zero
- positive values do not come from a truncated distribution

Two-part models

Two-part models are a more efficient way to handle situations in which we need to model at the same time two different processes underlying a certain phenomenon of interest, which is characterized by the fact that

- some event occurs (the binary part, i.e., having some expenditure, being working), and
- if the event occurs, some positive values are observed (the continuous part, i.e., the value of expenditure, the number of working hours)

Two-part models

We can formulate the case by thinking about two different latent processes Y_1^* and Y_2^*, and the variables we are actually able to observe are

$$y_1 = \begin{cases} 1 & \textit{if} \quad y_1^* > 0 \\ 0 & \textit{if} \quad y_1^* \leq 0 \end{cases}$$

and

$$y_2 = \begin{cases} y_2^* & \textit{if} \quad y_1^* > 0 \\ 0 & \textit{if} \quad y_1^* \leq 0 \end{cases}$$

Two-part models

In these situations, there are typically some variables that could serve to explain the expected value of y_2 and that are missing when $y_1 = 0$, for example

- satisfaction with job, and
- wage

Two-part models

Possible approaches could be

- to work on the full sample and recode missing values (i.e., assigning a value equal to zero)
- to use two separate models, one to model the probability, and the other to model the continuous observed variable

Two-part models

Possible approaches could be

- to work on the full sample and recode missing values (i.e., assigning a value equal to zero)

..By doing so, we essentially assume there is no difference in the effect of the independent variables for those who experience the event and those who do not, i.e., the effect of satisfaction with job on hours of work is the same for those working and those not working!!!

Two-part models

Possible approaches could be

- to use two separate models, one to model the probability, and the other to model the continuous observed variable

..By doing so, we need to work on a subsample, i.e., the effect of satisfaction with job on hours of work will be evaluated only in the subgroup of workers, and as a drawback, we may "lose" power, with a possible increase of variance

Two-part models

A more efficient way to solve the problem is to consider a two-part model in which we combine a probit or logit model for y_1, with a continuous/discrete (i.e., Gamma, Poisson) for y_2

Two-part models

The approach resemble what we have seen for Tobit regression and we already introduced the "estimation" problem for those models, which is solved by expressing the likelihood as the product of the desity functions f_1 when $y_1 = 0$ and f_2 when $y_1 = 1$, weighetd for the probability of $y_1 = c$ and using the ML method

Two-part models

Let's discuss the interpretation of parameters and the advantages of using this approach as compared to possible and "suboptimal" alternatives

Two-part models

Let's consider a practical problem and a study in which we assessed a continuous dependent variable of interest Y (i.e., hours of work in a week), and we are interested in evaluating the effect of **age** and **satisfaction with work** on the dependent variable having data on an adult population in which some of the units are currently not working, for the latter age could be assessed but not satisfaction with work

Two-part models

Assume to split the problem, specifying two different models, one for those working (W)

$$Y = \beta_{0,W} + \beta_1 AGE + \beta_2 SAT + \epsilon$$

and the other for those not working ($\bar{W}$)

$$Y = \beta_{0,\bar{W}} + \beta_1 AGE + \epsilon$$

with $\epsilon \sim N(0, \sigma^2)$

Two-part models

If we consider the observed variable Y being generated by two underlying latent variables Y_1^* and Y_2^* describing respectively

- working condition (0,1)
- working hours, if workers

Two-part models

We could rewrite the model for Y using a single equation

$$Y = \beta_0 + \beta_1 AGE + \beta_2 WORK + \beta_3 SAT^* + \epsilon$$

that is

$$Y = \begin{cases} \beta_{0,W} + \beta_1 AGE + \beta_3 SAT + \epsilon \quad if \quad WORK = 1 \\ \beta_{0,\bar{W}} + \beta_1 AGE + \epsilon \quad if \quad WORK = 0 \end{cases}$$

where we assume a similar residual error for workers and non-workers

Two-part models

Thus

- $\beta_{0,\bar{W}}$ is the expected value of Y for non workers at the average of age
- $\beta_{0,W} = \beta_0 + \beta_2$ is the expected value of Y for workers at the average of age
- β_1 is the expected difference in Y for a unit increase of age, holding constant working condition and satisfaction with work (if relevant)
- $\beta_2 = \beta_{0,W} + \beta_{0,\bar{W}}$ is the expected value of Y for workers minus the expected value of Y for non-workers
- β_3 is the expected difference in Y for a unit increase of satisfaction, only for workers

Two-part models

Two-part models could be extended to handle phenomena discussed before could also be extended to handle the analysis of longitudinal (correlated) data, in this setting the trade-ff between using an optimal approach and a "convenient and easy" solution may became more relavant, because, for example if we choose to model the binary and continuos part splitting the problem and working on subsample we may need to have multiple subsample

Two-part models

In a longitudinal framework, the two different latent processes Y_{1j}^* and Y_{2j}^* introduced before are also indexed by time T (j=1,...,t) and the variables we are able to observe are

$$y_{1j} = \begin{cases} 1 & if \quad y_{1j}^* > 0 \\ 0 & if \quad y_{1j}^* \leq 0 \end{cases}$$

and

$$y_{2j} = \begin{cases} y_{2j}^* & if \quad y_{1j}^* > 0 \\ 0 \quad if & y_{1j}^* \leq 0 \end{cases}$$

Two-part models

If we assume Y^*_{1j} follow a random effect logistic regression model

$$logit[P(y_{1j} = 1|x_j, U)] = x_{ij}\theta + U_i$$

where $i = 1, .., N$ denote the units and U_i is the individual-level random intercept, and if we assume y_{2j} is approximately normally distributed with a subject-time specific mean and constant variance

$$y_{2j}|y_{2j} > 0, x^*_j = x^*_{ij}\beta + V_i + \epsilon_{ij}$$

where V_i is the individual-level random intercept, and $\epsilon_{ij} \sim N(0, \sigma^2)$

Tobit and two-part models

- the Tobit model assumes that the same variables that explain the intensity of the observed variable y, also explain $y = 0$
- this is not true in the two-part model, where we could argue about two different processes generating the binary and the continuous part with possibly different independent variables behind, or even different effects of the same independent variables
- the Tobit model is sensitive to heteroskedasticity, which could be handled with two-part models

Finite-mixture models

There are other interesting applications where we "combine" different random processes to describe a phenomenon, which is known as **finite mixture models**

Finite-mixture models

Finite mixture models are typically used to solve clustering problems, where we assume that there are different underlying distributions of the same form that describe the data

Finite-mixture models

Finite mixture models are formulated considering that data are generated by a random X process that could be expressed as a density obtained as a weighted mean of densities

$$f(x) = \sum_{g=1}^{G} \pi_g f_g(x, \theta_g)$$

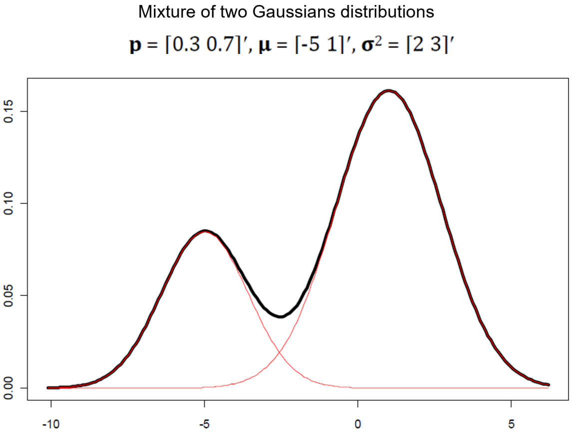
where $f_g(x)$ are density probability functions which could be of different forms (i.e., Normal, Beta, Poisson), and $\pi_g \geq 0$ with $\sum_g \pi_g = 1$ are weights (i.e., probabilities)

Finite-mixture models

Finite mixtures belong to unsupervised clustering techniques, which assume that different possible groups G in a population come from specific densities with specific parameters θ_g

Finite-mixture models

Example of finite mixture of Gaussian densities



Finite-mixture models

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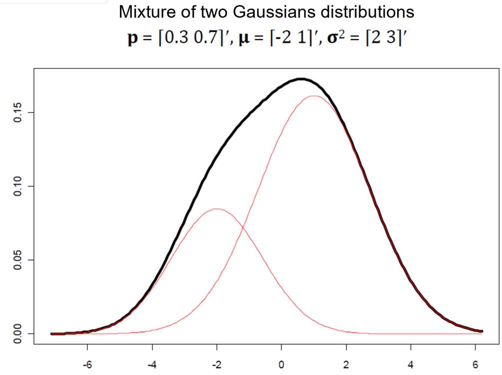


Finite-mixture models

In the case of fiinite mixture of Gaussian distributions the random variable X, has expectation

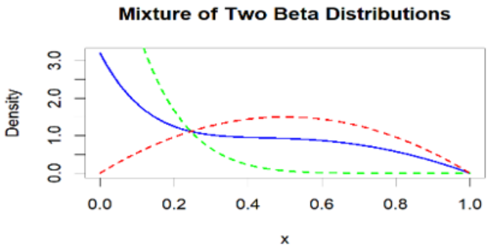
$$E(x) = \sum_{g=1}^{G} = \pi_g \mu_g$$

with μ_g being the expectation of the g component, as

$$E(x) = \int x\, f(x)\, dx = \int \sum_{g=1}^{G} \pi_g x\, f(x)_g\, dx = \sum_{g=1}^{G} \pi_g \int x\, f(x)_g\, dx$$

Finite-mixture models

Example of finite mixture of Beta densities



Finite-mixture models

Parameters estimation is obtained through ML, and the number of components can be chosen by using different approaches, one of the common use approach is to use a selection criterion based on the Akaike Information (AIC) or the Bayesian Information (BIC) criteria

Finite-mixture models

Once estimates are obtained, it is possible to estimate the posterior probability p_{gi} that the subject n comes from the component g

$$p_{gi} = \pi_g f_g(x, \theta_g) / \sum_{g=1}^{G} \pi_g f_g(x, \theta_g)$$